🛠 MEC 427 Advanced Sensors and Actuators Challenges 1-5

📅 Date & Time

- **Report Date:** `11-15-2025`
- **Time Block:** `11-10 - 11-15`
- **Time Spent This Session:** `3 hours / 30 minutes`
- **Cumulative Time on Project:** `25 hours / 0 minutes`

🗂 Project Overview

- **Project Title:** `Challenges 4 & 5: Gantry motion control system and Bead Sorting without controls perspective.`
- **Project Objective(s):**
- `Design an actuator system that can traverse a steel tension cable.`
- `Conseptualize a bead sorting system that uses sensors and actuators to sort beads without using a controls perspective.`
- **Current Phase:** `Implementation Phase`

🔧 Task / Activity Details

- **Module / Subsystem:** `Challenge 4 Gantry motion control system design`
- **Specific Task or Activity:**
- **Time Spent on This Task:** `1 hours / 30 minutes`
- **% of Total Time:** `50%`
- **Resources Used:**
  - **Hardware:**
  - **Software:**
  - **Tools:** `Grok AI, Tinkercad, Internet research.`
  - **Documentation:** `Design Sketches and AI generated STL.`

- **Module / Subsystem:** `Challenge 5 Bead Sorting without controls perspective`

- **Specific Task or Activity:**

- **Time Spent on This Task:** `2 hours / 0 minutes`

- **% of Total Time:** `50%`

- **Resources Used:**

  - **Hardware:**
  - **Software:**

- **Tools:** `Grok AI, Knowledge course resources, Internet research.`
- **Documentation:** `Notes, sketches, AI conversation.`

- ## Project Status Update:

- **Task/Module:** `Challenge 4 Gantry motion control system design`
- **Project Completion Status:** `70% complete`
- **Milestones Reached:** `Completed a conceptual design and prototype for gantry motion control system.`
- **Blockers or Issues:** `A dentist appointment caused me to miss a comp-team session`
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- **Task/Module:** `Challenge 5 Bead Sorting without controls perspective`
- **Project Completion Status:** `100% complete`
- **Milestones Reached:** `Completed a conceptual design for bead sorting system.`
- **Blockers or Issues:** `Not thinking about controls made it challenging to conceptualize the sorting mechanism.`

📦 Results & Deliverables

- **What Was Achieved:** `Completed Challenge 5 and made significant progress on Challenge 4.`
- **Artifacts Created:** `Conseptusual designs and AI generated STL files for gantry system. Sketches and notes for bead sorting system.`
- **Validation / Testing Outcomes:** `N/A`
- **Changes Made to Design or Approach:** `GroK AI was used to assist in design ideation and STL generation with tinkercad.`

📜 Narrative & Observations

> *Use this section for detailed descriptions, reflections, and context. Include technical notes, emotional observations, and rationale for decisions.*

- **Narrative Summary:** `Challenge 2: I didn't consider mounting the sensors on the water tower during my initial design, I compensated by using the CPX kit's alligator clips as extension wires and fixed them to the legs of the tower using tape. The sensor probes hang down into the lower container to detect water level. Callibration was a learning process, assisted by Copilot AI. The CPX capacitive touch sensors are rather sensitive and the wire insulation is suboptimal. An issue where the sensor readings would reverse polarity when the probes touched water was resolved by implementing a dual-threshold detection system, accounting for both increases and decreases in capacitance. This approach improved reliability` significantly. The voting system was implemented successfully, allowing the system to function correctly even if one sensor failed or provided erroneous readings.

- **Narrative Summary:** Challenge 4: I conceptualized a Tri-wheel gantry motion control system that uses a motor to move an actuator along a steel tension cable. I used Grok AI to generate design sketches and STL files for the gantry components. The design includes a motor, pulley system, and guide rails to ensure smooth movement along the cable.
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    - **Narrative Summary:** Challenge 5: A design session with Grok AI and my notes from the knowledge course. I conceptualized a bead sorting system where the blue and yellow beads sorted by color on a single track that splits into two paths. The beads pass through photoelectric sensors that detect their color. When a blue bead is detected, a small servo actuator diverts it to the left path, while yellow beads continue straight. This design avoids complex control systems by using simple sensor-actuator pairs for sorting. This consept can be scaled up to include more colors and sorting paths or multiple tracks for volume sorting.
